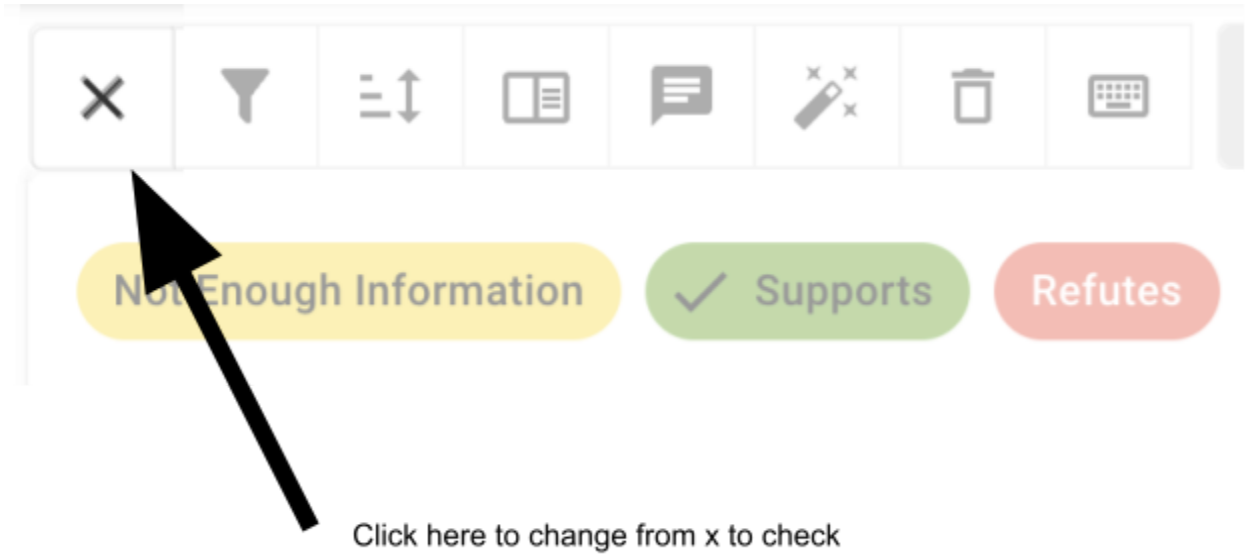
Guideline

The purpose of the exercise is to verify labels in our claim verification dataset. The examples presented to users will contain a claim (similar to a CIViC evidence item) and evidence (generally these are selections a user has highlighted from the related paper). The user must give the claim/evidence pair one of 3 labels

- **SUPPORTS**: all elements of the claim are supported by the evidence
- **REFUTES**: any element of the claim is refuted by the evidence.
    - Generally, this only applies to the significance term since other elements are not refutable
- **NEI**: There is not enough information to either refute or support the claim. A comment should be added to indicate what made this example NEI

Using Doccano

It is important that users do not discuss examples amongst themselves, this should be completed as a solo activity as one of the purposes will be calculating the inter-annotator agreement



To show progress in the progress bar in the doccano app (this app) users must click the "checked" button at the top left of each example after selecting a label. This is meant as a "double-checking" feature. Once clicked it will change from "x" to a checkmark and the progress count will increase by one

Examples

note, all of the examples below are made up content for the purposes of this exercise only. Gene names, variants, diseases etc were randomly selected and then sentences were generated to pair with them

Predictive Example

The following example would be labelled refutes because the evidence infers this variant causes resistance to the drug not sensitivity. Since sensitivity and resistance are mutually exclusive we label the example refutes.

> Claim
>
> | | |
> |---|---|
> | **molecular profile** | MTAP R122K |
> | **evidence type** | PREDICTIVE |
> | **significance** | sensitivity |
> | **therapies** | paclitaxel |
> | **disease** | breast cancer |
>
> Evidence
>
> In a cohort of 72 breast cancer patients, 12 were found to have a mutation in MTAP Arg122Lys. Patients with mutant MTAP demonstrated short overall survival (OS=4.4m) when compared to wildtype MTAP (OS=9.8m, p=0.032) after 4 courses of paclitaxel.

However, if the disease name were removed and the evidence now read

> In a cohort of 72 patients, 12 were found to have a mutation in MTAP Arg122Lys. Patients with mutant MTAP demonstrated short overall survival (OS=4.4m) when compared to wildtype MTAP (OS=9.8m, p=0.032) after 4 courses of paclitaxel.

We would have to give this example a label of NEI. Because, although the variant is clearly linked to resistance and not sensitivity, we have no way of knowing in what context. From just the selection of evidence provided it is possible this paper is talking about a different disease altogether and that this variant is only linked to resistance when paired with other variants characteristic to whichever disease they are studying

FAQ

Does the “additional information” table count as evidence

no, the additional information table is only present to aid curators in resolving alias terms without having to google them. If, for example, an element from the claim is present in the table but not in the evidence, that element is considered missing

What if the evidence doesn't make sense

All evidence is collected from scientific articles, while it may not make sense without additional context I would first try re-reading it critically, then if the evidence still does not make sense the user must use their best judgment on which label to apply. Regardless of which label is chosen, please add a comment to the example

What if the claim/evidence is more/less specific

In most cases this will result in an NEI example depending on how far off the specificity is. If it is very close SUPPORTS/REFUTES labels may be chosen at the users discretion but again, when there is any ambiguity, I would make a comment on the example before proceeding